Sahaj Bhadja

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Summary

MSCS candidate at USC with 2+ years building **Python** backend services, partner **APIs**, and **LLM** platforms for data ingestion, synchronization pipelines, and **AI-assisted development** in production Agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built ETL-style data ingestion pipelines with **FastAPI** and PostgreSQL for 5K+ monthly jobs with reconciliation, error handling, and **CI/CD** releases at 99%+ production reliability.
- Restructured async **LLM orchestration** with Python background workers and job-state transitions, processing 1K+ AI jobs with failure isolation and **AI-assisted** debugging, documentation, and developer productivity.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered production **Flask** backend **APIs** with **Claude** and multimodal LLMs for 150+ users, handling 10K+ daily requests with secure ingestion, validation, and stable peak-load performance.
- Participated in code reviews and design discussions using **AI-assisted development** for documentation and debugging, reducing feedback turnaround **60%** while maintaining stable production API performance.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **REST API** integrations and **AWS Glue** ETL pipelines for data synchronization, improving response times **30%** through query optimization and error-resilient ingestion workflows.
- Maintained Jenkins **CI/CD** with 20+ automated test suites in Agile sprints, boosting pipeline efficiency **68%** while diagnosing production issues through logging and validation reports.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed realtime **WebSocket** synchronization and **REST APIs** across eight microservices, syncing 2M+ records with low-latency updates and error handling for distributed partner inventory workflows.
- Shipped Docker and Jenkins **CI/CD** releases with validation gates, cutting deployment time **50%** while improving observability through logging, monitoring hooks, and production defect triage.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask APIs** and webhook payment flows for 50K+ records with sub-200ms responses, idempotent event handling, and cross-platform synchronization across 12+ integration points.
- Diagnosed production issues via logging and monitoring, reducing manual handling **70%** while documenting API behavior and collaborating on ambiguous integration requirements.

Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, Bash
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, partner integrations, microservices, API design
Data & AI: PostgreSQL, ETL pipelines, data ingestion, LLM systems, Claude, GenAI tooling, query tuning
Cloud & Ops: Distributed systems, event-driven architecture, WebSockets, AWS, Azure, Docker, Kubernetes, CI/CD
Practices: Logging, monitoring, code reviews, Agile, documentation, system design, automated testing

Projects

FinNexus Live | *Next.js, TypeScript, WebSockets, Kubernetes*
- Built a platform with partner APIs and WebSockets for 50K+ users and 1M+ transactions, real-time updates, secure access controls, and Kubernetes deployment automation.

Big-Brain Live | *PostgreSQL, LLM, REST APIs, GCP*
- Built an LLM ingestion and semantic search platform over 50K+ documents with PostgreSQL persistence, sub-second retrieval, and AI-assisted validation for production workflows.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker*
- Extended a Java and TypeScript SQL workspace supporting 10+ databases with schema-aware APIs, Docker deployments, and AI-assisted query generation tooling.